

R Practice Project 6 – Penguin Dataset Analysis

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ALY 6000 Introduction to Analytics

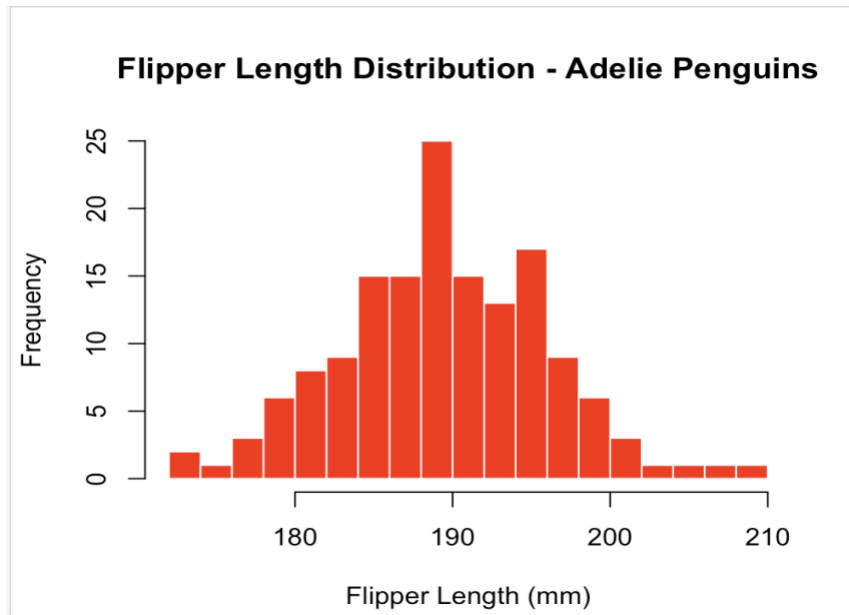
Professor Steward Huang

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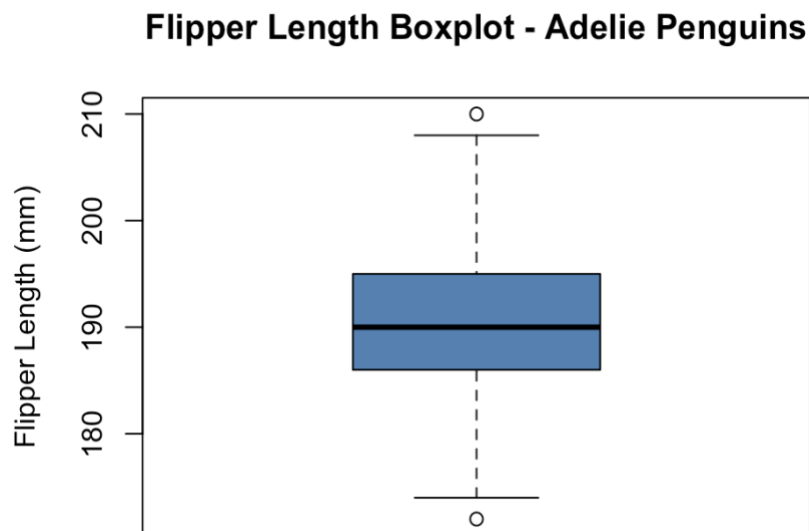
The purpose of ALY6000 Module 6 project was to analyze the distribution of flipper length of the adélie penguin and also explore the relationship between the flipper length and beak depth of the gentoo penguin. The process started with the typical data cleaning and preparation techniques. This involved installing and loading the tidyverse, janitor, lubridate, and skimr packages to ensure the being imported was formatted correctly and ready for a full analysis. The “palmerpenguins” package was installed and loaded into R Studio in order to start the analysis of flipper length for the adélie penguin and analysis of flipper length/beak depth of the gentoo penguin. After the package was installed and loaded, the flipper length of specifically adélie penguins was filtered out. The mean was determined to be 189.9536 millimeters, with a standard deviation of 6.5395 millimeters, and a variance of 42.7645 millimeters. The mean is simply the average which means the sum of all values divided by the number of values – so, on average, an adélie penguins’ flippers measure at right under 190 millimeters. The standard deviation of the adélie penguins’ flippers shows how spread out the values are from the average and the typical distance each value is from the mean. This essentially means that a large percentage of the data in a normal distribution for these penguins should have flipper length that varies by no more than 6.5 millimeters away from the average of 189.95 millimeters.

The adélie penguin data likely follows the normal distribution which shows a bell-shaped curve and has a symmetric distribution with few outliers, but none significant enough to skew the data. When analyzing the flipper length’s distribution, a histogram was created which showed a very normal distribution with a few outliers outside of the 183 – 195-millimeter range. It is clearly shown that the value right before 190 millimeters (so likely 189 millimeters) is the median because it is the most common flipper length. As the data starts to move away from this value, into values that are one-to-three standard deviations away from that mean value of 189.95,

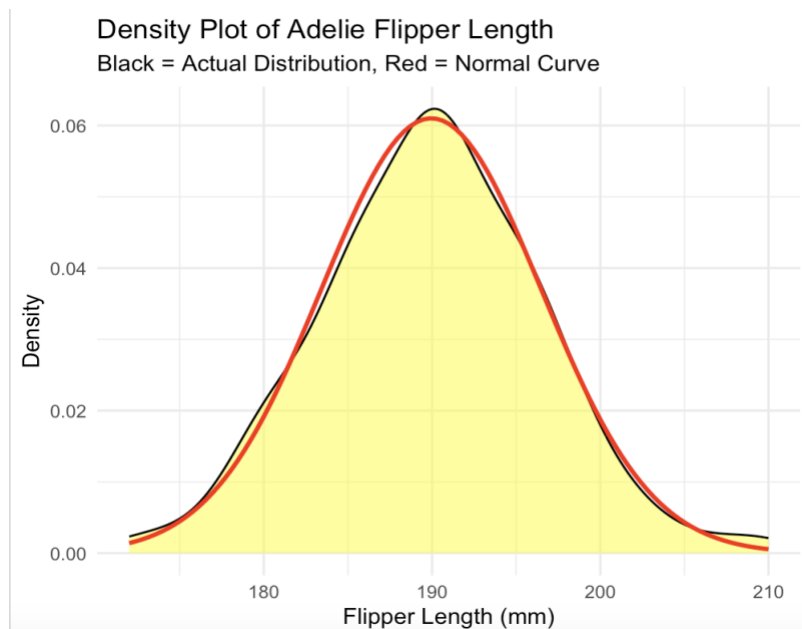
we see the frequency of this occurring as much less. There are values outside of the 175 millimeter and 200-millimeter range, but they occur very rarely and have minimal impact on the rest of the data.



A boxplot was also created which shows that the data is very clustered with a very small IQR and range overall. In this boxplot, there are outliers that are shown falling outside of the upper and lower whisker, but they are not very common.



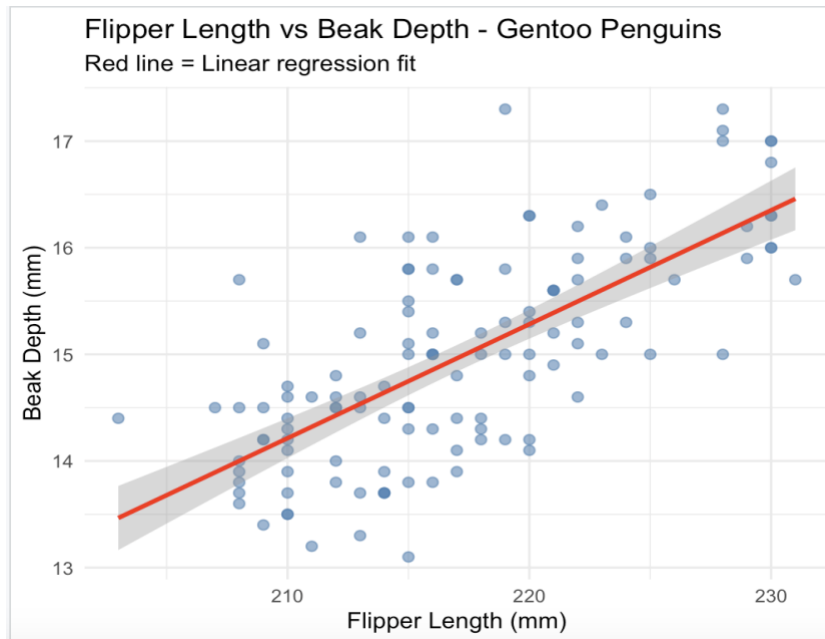
The fact that the lower and upper quartile are both very close to each other shows a distribution similar to one that is symmetric with no significant outliers. Furthermore, the median being almost split down the middle of the interquartile range is a good indicator that the data in the adélie penguin data set is following a normal distribution with a bell-shaped curve.



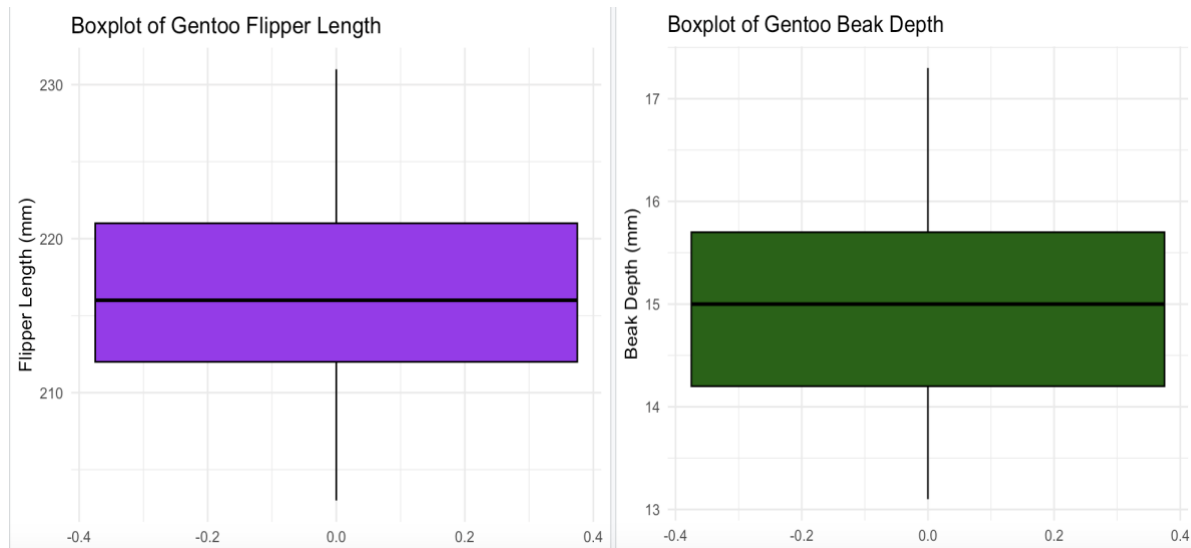
The density plot of adélie flipper length shows the shape of the actual distribution compared to a normal distribution with the x-axis representing flipper length and the y-axis representing density. It is pretty clear that the distribution shown in black for the adélie penguin flipper length has a distribution that is nearly symmetrical and follows the normal distribution which can be shown in red in the chart. All three of these plots are a good indication that adélie penguin flipper length follows a distribution that is similar to one that is normal with no significant outliers impacting the data. Lastly, a Shapiro-Wilk test was run through the code to statistically confirm a normal distribution.

In order to explore the relationship between flipper length and beak depth of the gentoo penguin, the data on both those variables for these gentoo penguins was specifically filtered out

to be analyzed. The relationship between flipper length and beak depth of the gentoo penguin can be stated to be that as flipper length of these penguins increases, so does beak depth.



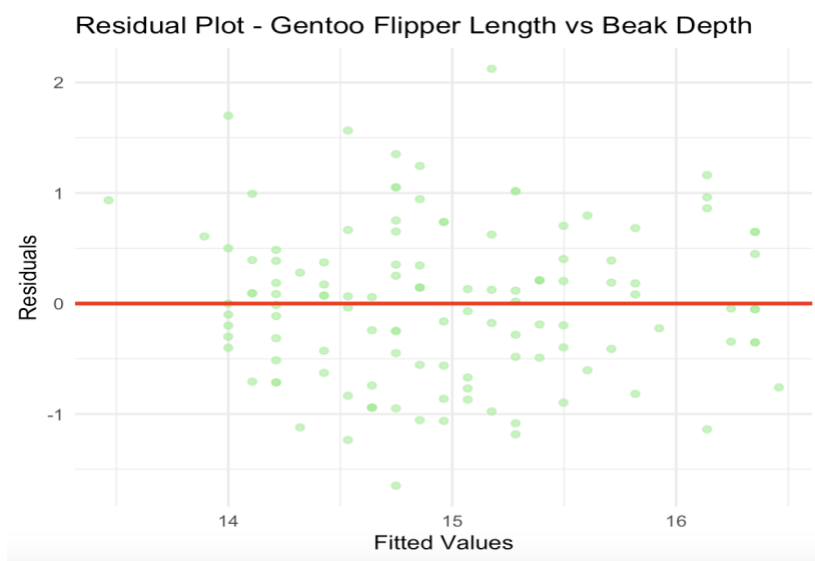
A correlation test was run through the R Studio platform which confirmed gentoo penguin's beak depth and flipper length to have a strong positive correlation at $r = 0.71$ for this relationship. The line plot shows the different penguins listed in the steel blue color with very few outliers. The red line demonstrates a linear model, and it can be seen that as the flipper length increases for these penguins with the larger flippers, their beak depth also tends to increase in what is similar to a linear relationship. It should also be noted that flipper length vs beak depth was different by sex with males tending to have a larger beak depth by approximately 1.5 millimeters and flipper length that is 8-10 millimeters longer compared to female gentoo penguins.



The boxplots of both gentoo flipper length and beak depth show that the distribution is very normal, with the data for flipper length ranging from approximately 202 millimeters to 232 millimeters. The visual presentation of both the flipper length and beak depth box charts show that a large portion of the data is included within the IQR and close to the mean values. The gentoo beak depth box plot shows that data for beak depth ranges from about 13.2 millimeters to 17.5 millimeters. Since the data being presented is indicating a normal, symmetrical distribution, it can be interpreted that as flipper length increases so does beak depth for these gentoo penguins. The linear model for the gentoo penguins makes sense because the ones with larger flippers also have a greater beak depth.

A residual plot was generated to show the relationship between gentoo flipper length vs beak depth. The residual plot is aimed at showing the difference between the actual observed values and the predicted values from the regression model. There is no obvious pattern or curve in the residuals, and the points are scatter randomly about and below the red line – this helps confirm the linear relationship between flipper length and beak depth. The spread of the residuals shows that most fall between -1 and +1, with one potential outlier just above 2 towards the top of

the graph. This point is noticeably further from zero than others, but not enough to be statistically significant or invalidate the relationship between gentoo flipper length and beak depth.



The fitted values range from around 14 millimeters to 16 millimeters which aligns with our already-known mean beak depth of around 15 millimeters. This residual plot is essentially providing strong evidence and confirming that this is in fact a linear relationship for gentoo penguins' flipper length and beaks.

The analysis of adélie penguin flippers reveals a clear normal distribution with a mean of 189.95 and a standard deviation of 6.539. The histogram and box plots show that the data is mostly symmetrical with a bell-shaped curve. For the future, normal distribution based statistical tests should be used with adélie penguin data on flipper lengths and outliers that are more than 3 standard deviations away should continue to be monitored. The analysis of the gentoo penguin flipper length and beak depth also reveals a linear model with a scatterplot showing that as flipper length increases, so does beak depth. Additional predictors should be investigated to see if body mass, year, or island location impacts the beak depth or flipper length of these adélie or gentoo penguins.

References

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